

Notes on HeFESTo

The purpose of HeFESTo is to determine self-consistently the equilibrium phase assemblage and physical properties of multi-component mantle systems. Much of the foundation is given in *Stixrude & Lithgow-Bertelloni* (2005, 2011, 2022, 2024). Some applications are discussed in *Xu et al.* (2008) and *Stixrude & Lithgow-Bertelloni* (2012). When using HeFESTo, please cite *Stixrude & Lithgow-Bertelloni* (2005) or *Stixrude & Lithgow-Bertelloni* (2011), depending on the application, and, if separate, the paper specifying the parameter set used. *Stixrude & Lithgow-Bertelloni* (2024) specifies the latest published parameter set.

The code and parameter sets are freely available on github. The code:

<https://github.com/stixrude/HeFESToRepository>

The latest parameter set:

https://github.com/stixrude/HeFESTo_Parameters_010123

Briefly the code computes for a given pressure, temperature, and bulk composition, i) the stable phase assemblage, including the abundances and compositions of the phases coexisting in equilibrium ii) the physical properties of those phases iii) the physical properties of the multi-phase assemblage and iv) scaling of physical properties to seismological time scales. More details on capabilities are given in the specification of the input file below. Items marked (*) are capabilities that are still unpublished.

The input file is called:

control – specifies the bulk composition, range of conditions to be explored, and the set of mantle phases and species (end-members) to be included in the minimization.

The code also requires a set of files (parameter set) specifying the properties of each species.

There are many output files, including:

fort.56 – Physical properties of the assemblage

fort.66 – Relative amounts of phases

fort.99 – Absolute amounts of species

Further details on the structure of these files are given below.

References

Stixrude, L. and C. Lithgow-Bertelloni, Thermodynamics of mantle minerals: 1. Physical properties, *Geophysical Journal International*, 162, 610-632, doi: 10.1111/j.1365-246X.2005.02642.x , 2005.

- Stixrude, L. and C. Lithgow-Bertelloni, Influence of phase transformations on lateral heterogeneity and dynamics in Earth's mantle, *Earth and Planetary Science Letters*, 263, 45-55, doi: 10.1016/j.epsl.2007.08.027, 2007.
- Stixrude, L. and C. Lithgow-Bertelloni, Thermodynamics of mantle minerals II, Phase equilibria, *Geophysical Journal International*, 184, 1180-1213, 2011.
- Stixrude, L. and C. Lithgow-Bertelloni, Geophysics of chemical heterogeneity in the mantle, *Annual Reviews of Earth and Planetary Sciences*, 40, 569-595, doi: 10.1146/annurev.earth.36.031207.124244, 2012.
- Stixrude, L. and C. Lithgow-Bertelloni, Thermal expansivity, heat capacity, and bulk modulus of the mantle, *Geophysical Journal International*, 228, 1119-1149, 2022.
- Stixrude, L. and C. Lithgow-Bertelloni, Thermodynamics of mantle minerals – III. The role of iron, *Geophysical Journal International*, 237, 1699-1733, 2024.
- Xu, W., C. Lithgow-Bertelloni, L. Stixrude, and J. Ritsema, The effect of bulk composition and temperature on mantle seismic structure, *Earth and Planetary Science Letters*, 275, 70-79 doi:10.1016/j.epsl.2008.08.012, 2008.

Structure of control file.**- Line 1.**

First pressure (or density) (P1), last pressure (or density) (P2), number of pressure intervals (nP), first temperature (or entropy) (T1), last temperature (or entropy) (T2), number of temperature intervals (nT), ensemble flag (iensem), an additional temperature value (superad), frozen temperature (tfreeze).

Possible choices for the ensemble flag (iensem):

0: NPT ensemble. This is the most typical choice.

-1: read in a temperature profile from a file that must be called "ad.in" and which must contain columns: Pressure, Unused, Temperature. In this case, T1 and T2, are ignored and nT=0. The value of superad is added to the temperature as read in from the file.

-2: NPS ensemble*. In this case, T1 and T2 are entropies in units of J/g/K. The value of superad is an initial guess at the temperature at P1.

-3: Hugoniot*. The code computes the Hugoniot, with the unshocked state given by the equilibrium assemblage at P1, T1. T2 is ignored and nT=0.

-4: NVT ensemble*. In this case P1 and P2 are volumes in units of cm³/g.

-5: NVS ensemble*. In this case T1 and T2 are entropies in units of J/g/K and P1 and P2 are volumes in units of cm³/g.

-6: Frozen equilibria*. The equilibria computed at P1 and T1 are unchanged for the remainder of the pressure, temperature calculations.

- Line 2.

Number of components, c. Further entries on this line are not used.

- Line 3.

This line is not used. Do not delete.

- Lines 4 to 4+c-1.

Name of component, first amount of component, last amount of component, number of intervals. Note that the number of intervals should be the same for all components.

Name of component is specified by the chemical symbol of the cation (for homovalent oxide compositions) or atom.

For homovalent oxide compositions, amounts are specified in molar amount of the oxide on a single cation basis (e.g. SiO₂, AlO_{1.5}). (Note that HeFESTo uses absolute amounts, not molar fractions).

For compositions containing multiple valence states of iron, the amounts of each *atom* are specified (including O).

- Line 4+c

Level of theory (normally only for testing): ityp, ivtyp, ittyp, iltyp

Entries should be:

1,1,1,2 (for parameter set inv251010 and earlier)

1,1,1,1 (for later parameter sets)

ityp .eq. 1: Volume dependence of VDOS: full theory quadratic in theta

ityp .eq. 2: Volume dependence of VDOS: Full theory linear in theta

ityp .eq. 3: Volume dependence of VDOS: q=constant

ivtyp .eq. 1: Volume dependence of G: Full expansion

ivtyp .eq. 2: Volume dependence of G: Truncated expansion

ityp .eq. 1: Temperature dependence of G: Full theory

ityp .eq. 2: Temperature dependence of G: eta prop. V

ityp .eq. 3: Temperature dependence of G: eta constant

ityp .eq. 4: Temperature dependence of G: eta prop. Gq

iltyp .eq. 1: New Landau theory, post SLB11*

iltyp .eq. 2: Landau theory as in SLB11

- Line 4+c+1

Path to the directory containing the parameter files

- Line 4+c+2

Number of end-member species

- Remaining lines

These specify the phases to be included; a code to indicate whether they are present (1) or not (0) in the initial guess; and the names of the end-member species to be included for that phase. Note that the initial guess must span the bulk composition. Names of phases and species must match the names of the corresponding parameter files.

Note that because naming conventions have changed, for example because a phase has been discovered in nature since our first HeFESTo publications (e.g. bridgmanite), some incompatibilities have developed between phase and species names that must be used in the control file and those that are referred to in our more recent publications. Some of these correspondences are listed below,

giving first the name of the phase or species as it must appear in the control file, and second, the name of the phase of species as it is used in Stixrude & Lithgow-Bertelloni (2024)

c2c	:	hpcpx
mgc2	:	hpcen
fec2	:	hpcfz
pv	:	bg
xpv	:	xbg (where x=mg,fe,al,fe,hl,fa,cr)
mw	:	fp
il	:	ak
mgil	:	mgak
feil	:	feak

Structure of fort.56

- **Line 1.** Header echoing the parameter set used and the wall clock time.

- **Line 2.** Header specifying the contents of the columns:

Pressure (GPa), Depth (km), Temperature (K), Density (g/cm³), Bulk sound Velocity (km/s), Shear wave velocity (VS) (km/s), Longitudinal wave velocity (VP) (km/s), VS modified by attenuation (km/s), VP modified by attenuation (km/s), Enthalpy (kJ/g), Entropy (J/g/K), thermal expansivity (10⁻⁵ K⁻¹), isobaric heat capacity (J/g/K), isothermal bulk modulus (GPa), Shear attenuation (QS), Longitudinal attenuation (QP), Quenched density (g/cm³), most abundant phase.

- **Additional notes**

- The thermal expansivity, isobaric heat capacity, and isothermal bulk modulus include metamorphic contributions (Stixrude & Lithgow-Bertelloni, 2022). The wave velocities do not include metamorphic contributions.
- Our approach to estimating the quality factor Q is described in Stixrude & Lithgow-Bertelloni (2007).